



POORNIMA

INSTITUTE OF ENGINEERING & TECHNOLOGY

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Basic Mechanical Engineering

251ME3-10/252ME3-10

Notes

Unit-3

Application of Pumps

- (1) Reciprocating Pump
 - (2) Centrifugal Pump
 - (3) Compressor Pump
 - (4) Submersible pump
- } To pump water to overhead tank in building and also for agriculture
- (5) Vane pumps — To pump semi viscous liquid.
 - (6) Gear pumps — To pump viscous oil for lubrication
 - (7) Injector pumps — To pump boiler feed water during its operation
 - (8) Different types of fuel pumps — To pump petrol or diesel oil in I C engine
 - (9) Screw pump — To pump liquid mixed with solid material.

Reciprocating Pump :- classified as

- (1) Single Acting
- (2) Double Acting

Working of a Single Acting Pump (Reciprocating)

Parts :-

- (1) cylinder with a piston
- (2) Piston rod, connecting rod, crank and crankshaft
- (3) Suction pipe with foot valve and filter
- (4) Delivery pipe
- (5) Suction valve
- (6) Delivery valve

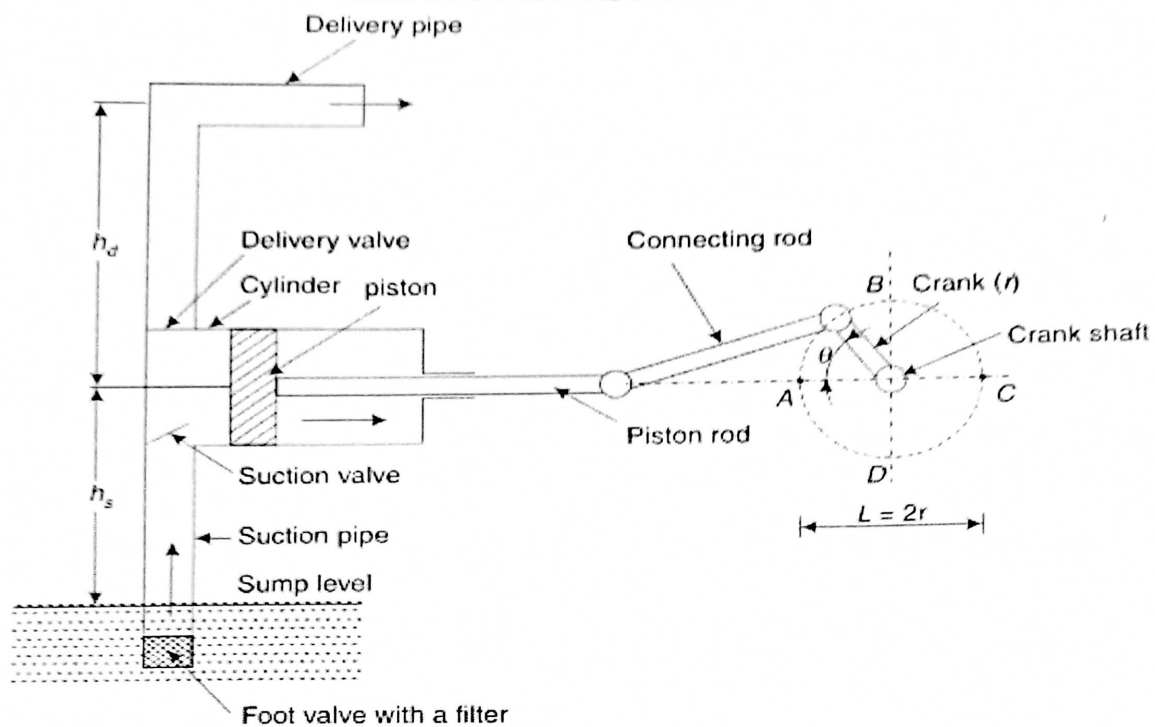
The crankshaft is coupled to an electric motor or diesel engine. When motor or engine started the piston moves to and fro inside the cylinder. When the piston moves right in the direction of arrow a vacuum will be produced in the cylinder due to which the suction valve opens and water is sucked up from the sump and enters the cylinder through suction pipe.

When the piston moves in left direction pressure is created in water due to which the delivery valve is opened and water is forced into the delivery pipe and finally to the required height.

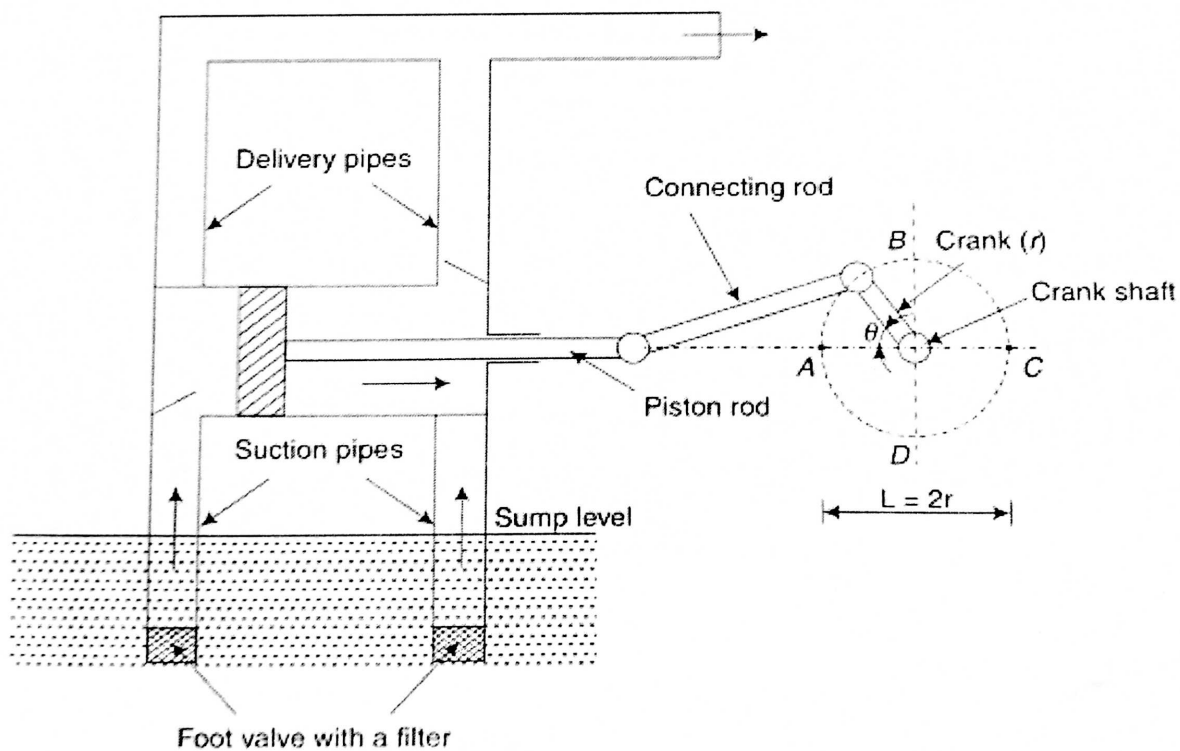
Working of Double Acting Pump

In the double acting pump water acts on both sides of piston. There are two suction pipes and two delivery pipes. When there is a suction stroke on one side of the piston there will be a delivery stroke on the other side of the piston. Thus for each revolution of the crankshaft there will be two delivery strokes and so double the amount of water is delivered by this pump.

Single Acting Reciprocating Pump



Double Acting Reciprocating Pump



Difference Between Single Acting and Double Acting Pumps

S ^{no}	Single Acting	Double Acting
(1)	One Suction Pipe	(1) Two suction Pipe
(2)	One delivery pipe	(2) Two delivery pipes
(3)	water act one side of the piston	(3) water acts both side of the piston
(4)	During one rotation of the shaft there is only one valve	(4) Two delivery strokes
(5)	water pumped will be less	(5) Double the amount
(6)	power of the motor or engine less	(6) Higher power required
(7)	cost is less	(7) Cost is more.

Advantage of Reciprocating pump

- (1) It can deliver the required flow rate very precisely
- (2) It can deliver fluid at very high pressure.
- (3) No priming is needed in Reciprocating pump.
- (4) Efficiency of reciprocating pump is 10% to 20% greater than the efficiency of Centrifugal pump.
- (5) It give continuous rate of discharge

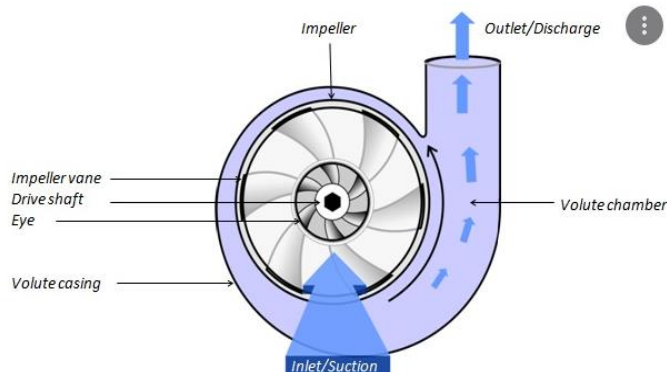
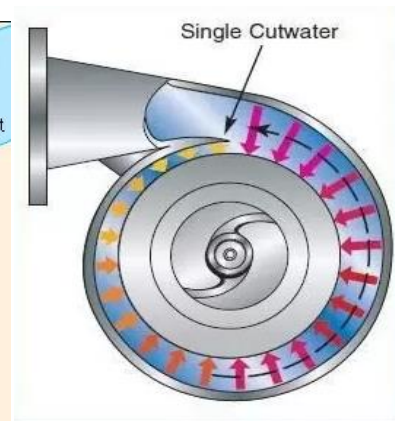
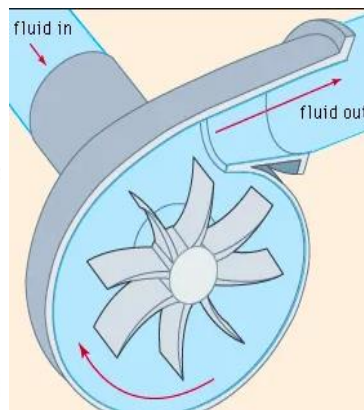
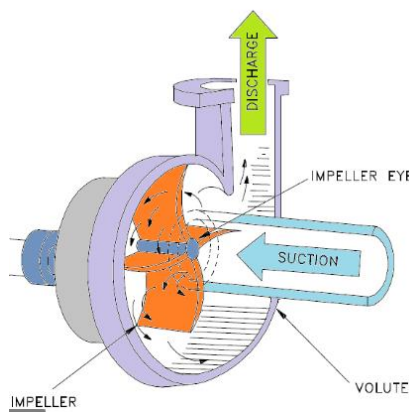
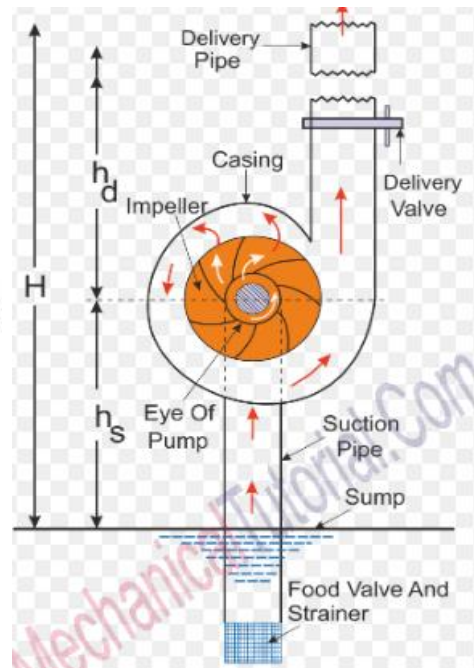
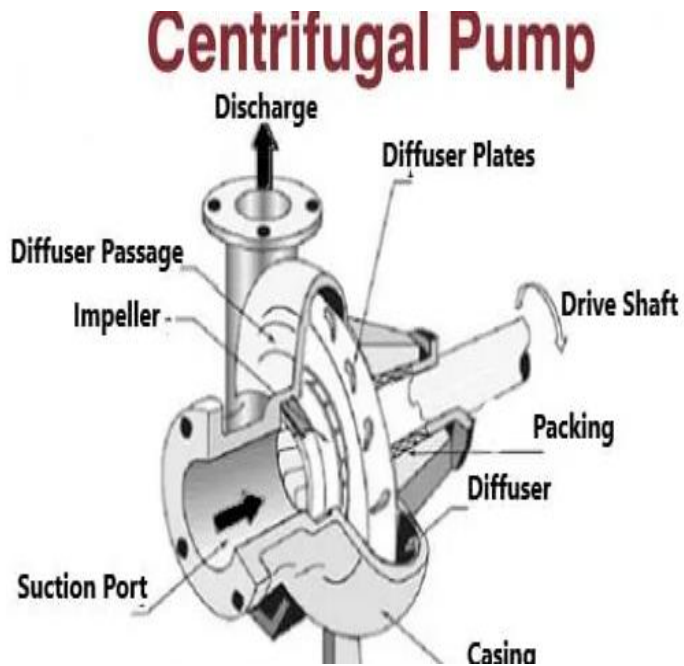
Disadvantages of Reciprocating pump

- (1) The pump is very costly (Capital cost is high)
- (2) Viscous fluid are difficult to pump with reciprocating pump
- (3) The maintenance cost of the pump is very high, as there is a large number of parts.
- (4) The flow rate is less.

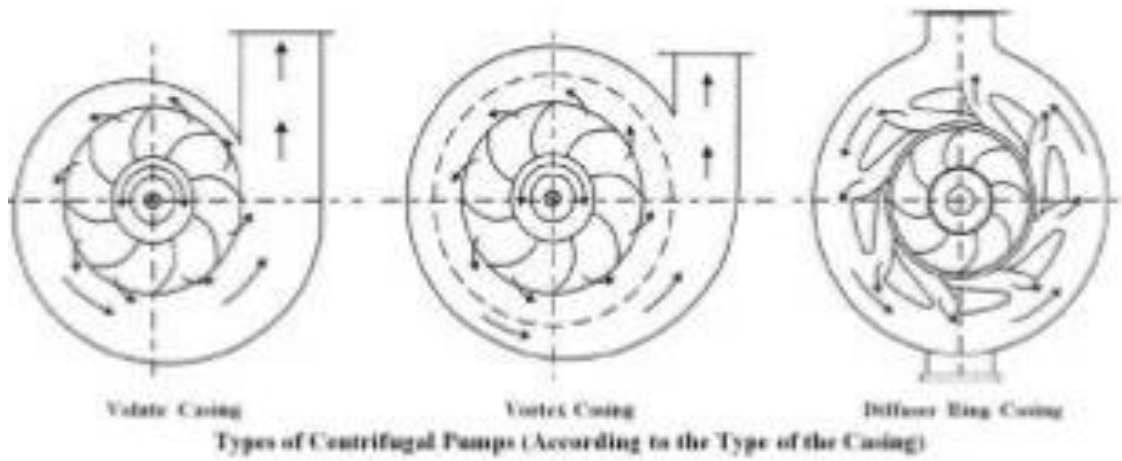
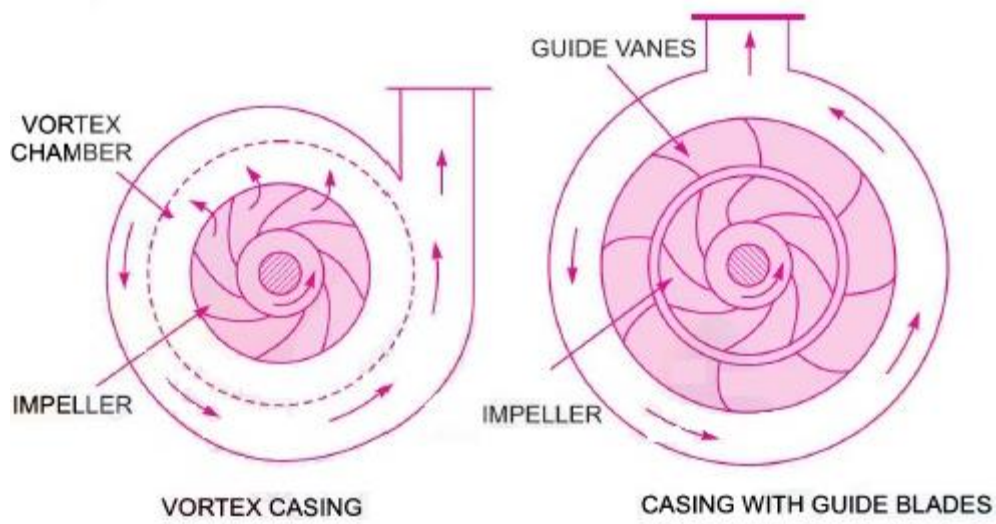
Application of Reciprocating pump.

- (1) It is used in vehicle washing center.
- (2) It is used in cycle pump football pump etc.
- (3) It is used in hydraulic jack.
- (4) It is continuously used in pneumatic pressure system.
- (5) It is used in gas industries.
- (6) It is also used in oil industry.
- (7) It is also used in petroleum industry.

Centrifugal Pumps



Volute type



Centrifugal Pump

(10)

Main part of Centrifugal Pump

- (1) Impeller
- (2) Casing
- (3) Suction pipe-fitted with a foot valve and filter
will not allow water to come down.
- (4) Delivery pipe

Impeller

The Impeller is a metallic disc fitted with number of curved vanes. Initially water will be poured inside the casing and process is called priming. So after priming the impeller will be immersed in water inside the casing. When the Impeller is rotated by an electric motor or an engine it will produce centrifugal force due to which kinetic energy or velocity energy will be produced in the water. This is finally converted into pressure energy due to which water is pumped.

Casing

The casing surround the Impeller. The area between the

Impeller and casing is gradually increasing till the delivery pipe. Due to this the velocity of flow of water will be gradually decreased. Due to the reduction in kinetic energy the pressure energy of water is increased. The following three types of Casing normally used:-

(1) Volute casing

(2) Vortex casing

(3) Casing with Guide Vanes or Diffuser casing.

(1) Volute casing

The gap between the Impeller and casing is comparatively less, so eddies or turbulence will be formed which reduce the efficiency of the pump. However the size of the casing will be small and the cost will be less.

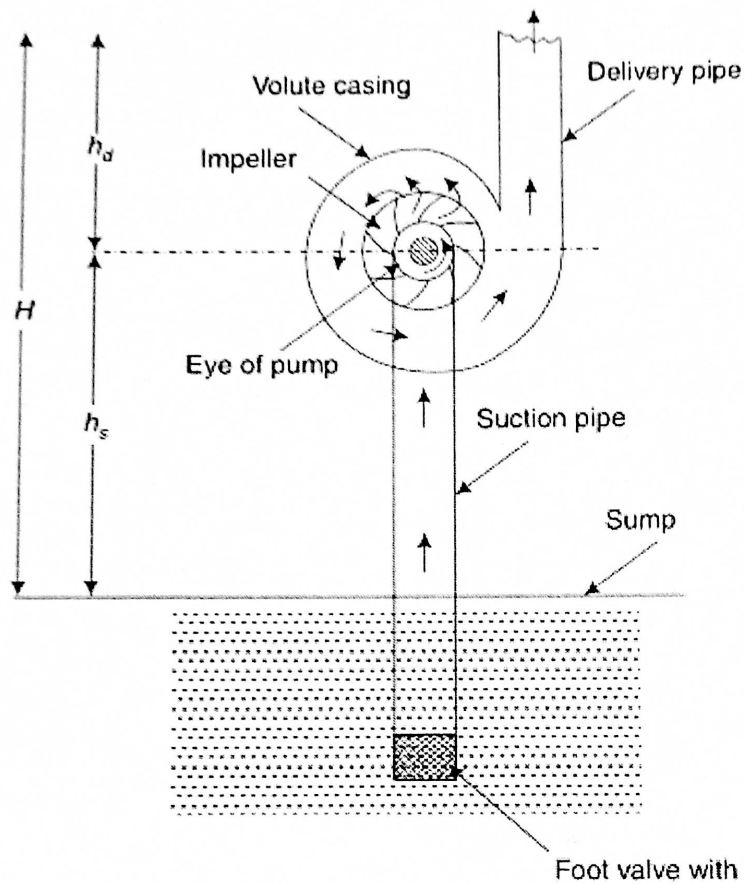
(2) Vortex casing

The space between the Impeller and casing is increased by introducing a circular chamber. As eddies and turbulence are reduced the efficiency of the pumps with Vortex casing is increased.

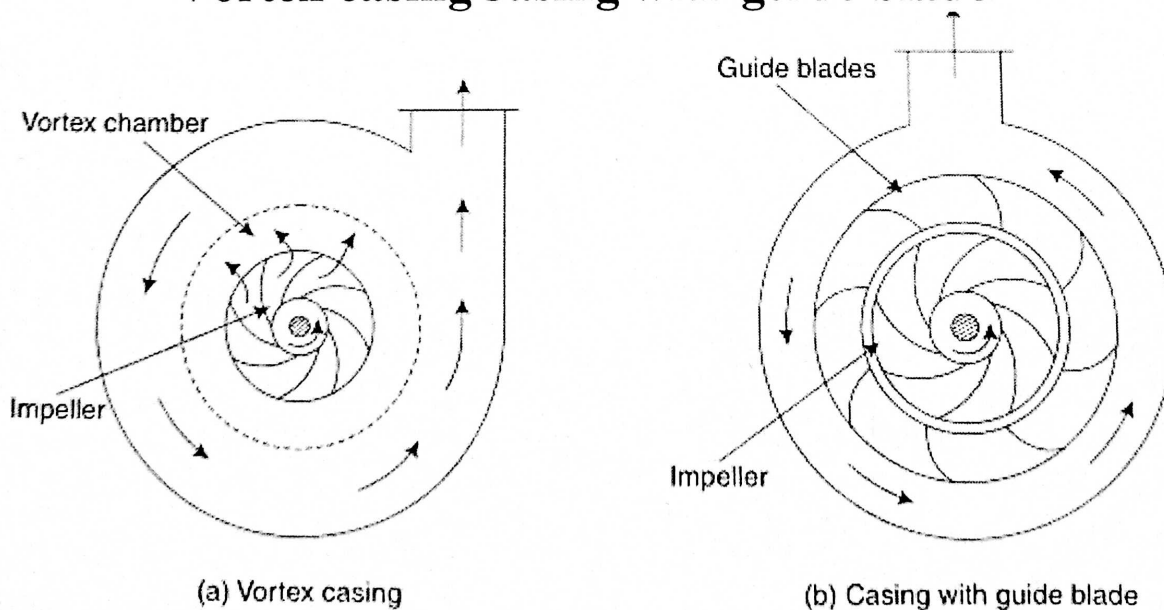
(3) Casing with Guide Vanes or Diffuser casing.

The Impeller is surrounded by a set of guide vanes or diffuser. The shape of guide vanes should be carefully designed to ensure that the water flow from the Impeller enter the guide vanes without shock. Gradual increase in area of guide vanes reduce the velocity of flow resulting in pressure increase.

Centrifugal Pump



Vortex casingCasing with guide blade



Difference between Reciprocating & Centrifugal pump

S ^{no}	Reciprocating Pump	Centrifugal Pump
(1)	Can be used for high head	(1) Medium Head
(2)	Can pump less quantity	(2) Large quantity can be pumped.
(3)	Discharge is not continuous	(3) Continuous flow.
(4)	Cost is high	(4) Cost is low
(5)	Cannot run at high speed	(5) Can run at high speed.
(6)	Cannot run or pump with viscous fluid	(6) Can pump viscous fluid.

Advantages of Centrifugal Pump

- (1) Centrifugal pump does not have leakage issue.
- (2) They are able to pump hazardous as well as sensitive fluid.
- (3) There is no problem of heat transfer as the space between the motor and chamber is sufficiently large.
- (4) There is no power loss due to friction and they are very simple in structure and easy in handling.

Disadvantage of Centrifugal pump

- (1) Manage resonance in centrifugal pump result in small loss of energy.
- (2) The risk of the clogging of pipe may arise due to particle attraction nature of magnetic drive.
- (3) Vibration due to surrounding atmosphere can damage their pump.
- (4) The risk of Cavitation is always there.

Application of Centrifugal pump

- (1) These pump are used in building in regular water supply.
- (2) They are used for fire protection related system.
- (3) Centrifugal pumps are used in pharmaceutical Industry.
- (4) They are also used coolant recirculation.
- (5) Used in refrigeration system to flow the refrigerants.
- (6) These pumps are used in sprinkling, Irrigation, drainage system.

Introduction

Refrigeration is defined as the science of providing and maintaining the temperature below the surrounding atmosphere.

In this heat has to be removed from the source at a lower temperature and rejected to the atmosphere at a higher temperature. Heat is flow in reverse direction the second law of thermodynamic stipulates that external work should be supplied. Therefore a refrigerator is a heat pump using power.

Unit of Refrigeration

Ton of Refrigeration (TR) which is defined as the quantity of the heat to be removed to produce one ton of ice at 0°C within 24 hours when initial condition of water is also 0°C .

$$1 \text{ TR} = 210 \text{ KJ/min or } 3.5 \text{ KW}.$$

Performance of Refrigerator

When a refrigerator is removing Q amount of heat consuming W amount of work then COP is Q/W

Application of Refrigeration

- (1) In water cooler to supply cold water for drinking.
- (2) To manufacture ice.
- (3) for preservation of food, vegetables, milk, icecream etc
- (4) for Preservation of perishable like fish, mutton, chicken etc
- (5) Preservation of medicines, blood, tissues etc in hospital
- (6) Preservation of dead bodies in mortuaries in hospital.
- (7) Different types of industrial applications.
- (8) for air conditioning in house, Hospital, office theatres, Computer center etc.

Refrigerants

A Refrigerant is a working fluid in a refrigerator. It is capable of absorbing heat at a lower temperature and rejecting the heat at a high temperature in the form of sensible heat or latent heat or both.

Desirable properties of Refrigerants

- (1) Low boiling, low freezing point, high latent heat of evaporation.
- (2) Low specific heat and low viscosity
- (3) It should be easy to liquefy.
- (4) Odourless and no hazardous effect on leakage.
- (5) Chemical stability.
- (6) Non flammable
- (7) Low cost

Types of Refrigerants

(1) Freon-12

This is called dichloro difluoro methane (CCl_2F_2). This is mostly used in domestic refrigerators, water cooler and freezers. It has a boiling point of -29°C .

(2) Freon-22

It is monochloro difluoro methane (CHClF_2) and it has a boiling point of -41°C . It is mostly used in air conditioners.

(3) Ammonia

It is mostly used in absorption system. It has a boiling point of -33.3°C .

(4) Other refrigerants include carbon dioxide (CO_2), sulphur dioxide (SO_2) and methyl chloride (CH_3Cl). CO_2 is mainly used in marine refrigerators.

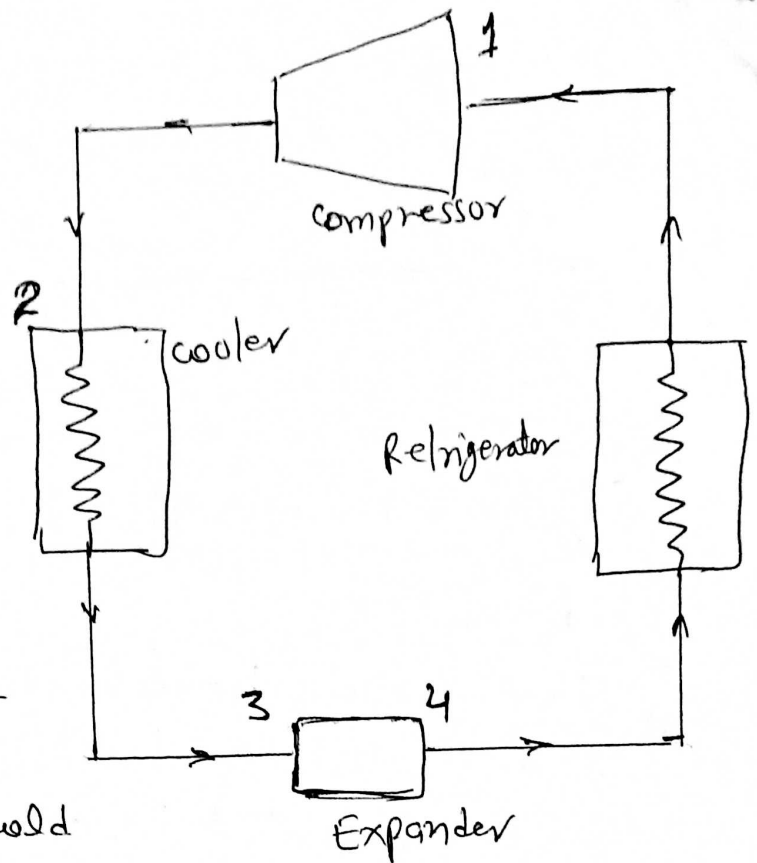
Air Refrigeration

Due to the availability of free air from atmosphere, air was used as a refrigerant even in commercial applications. But due to low heat capacity and low COP not used in modern system.

Closed Air Refrigeration cycle

In this cycle air is passed through pipes. Air is compressed in the compressor where the pressure and temperature are increased.

The warm air from ~~atmosphere~~ compressor is cooled in a cooler with water. The cold



air from cooler expands in the expander causing a big drop in pressure and temperature. Cold air from the expander is passed on to the refrigerator, where it absorbs the heat and makes it cold. Then the air returns to the compressor and the cycle is repeated.

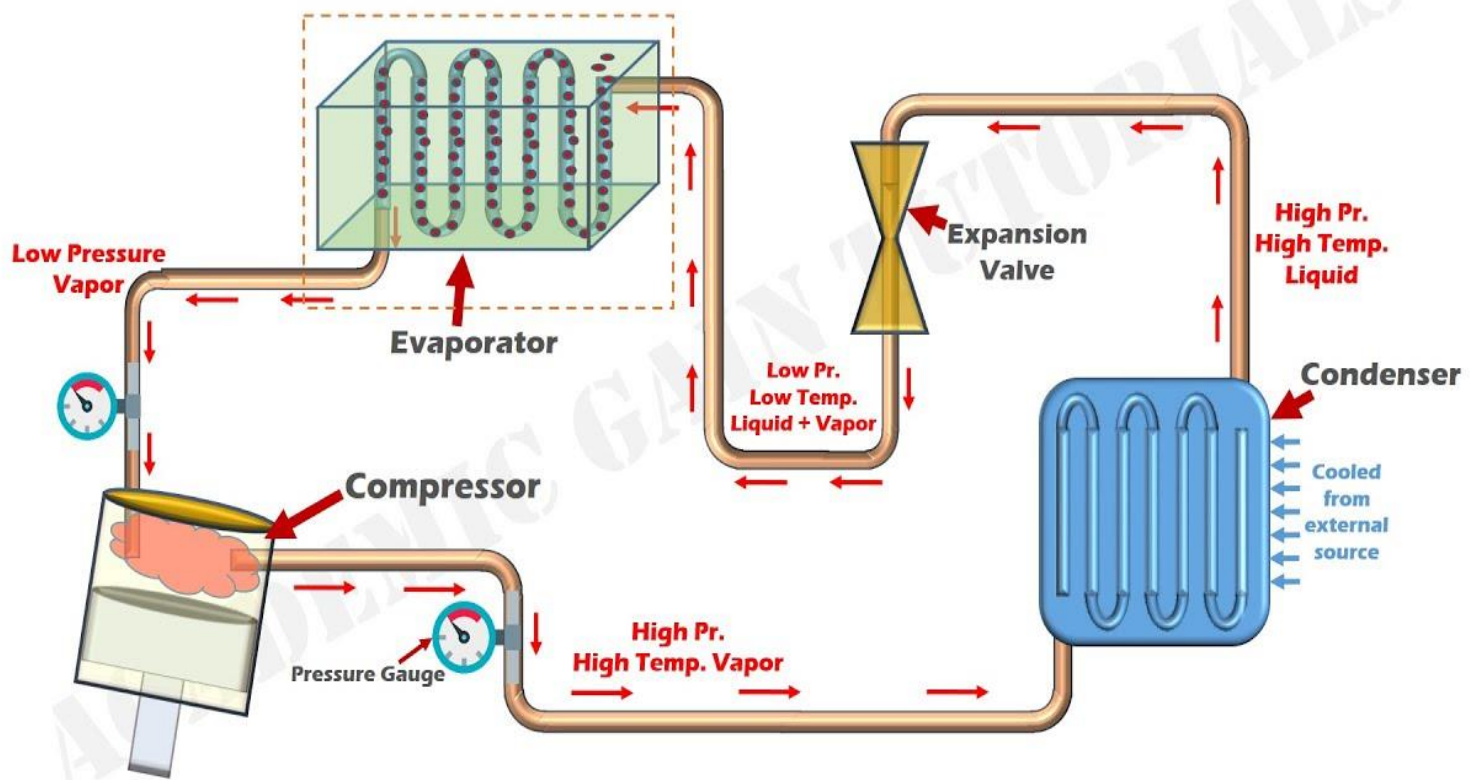
Advantages of air refrigeration

- (1) Air is easily available, (free of cost).
- (2) Leakage of air will not create any problem.
- (3) Air is very safe as it is non-toxic and non-flammable.
- (4) The air system is light in weight.
- (5) Maintenance cost is very low.
- (6) Charging of the refrigerant is very easy.

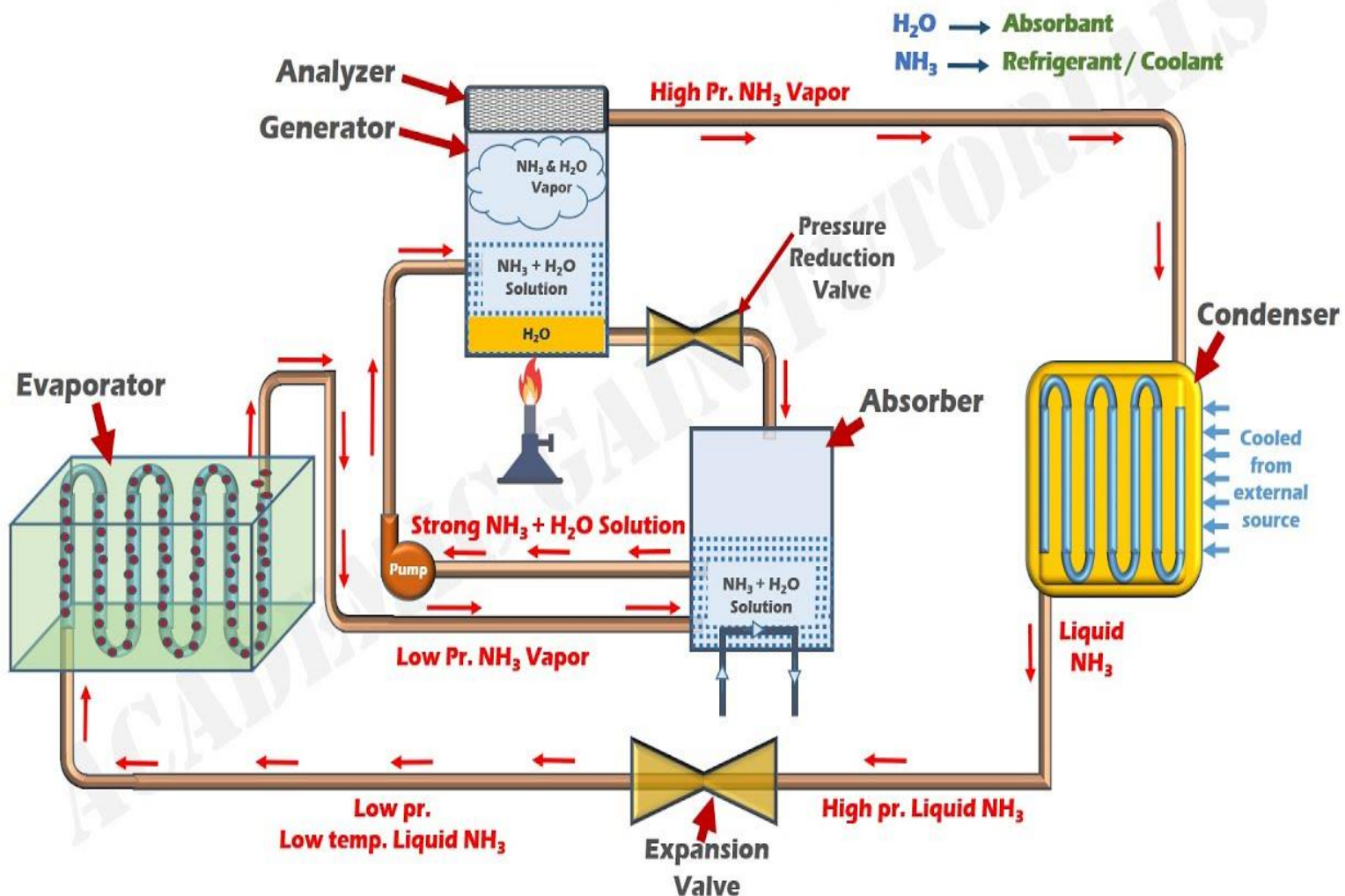
Disadvantage of refrigeration

- (1) COP is very low.
- (2) Low heat carrying capacity.

Vapour Compression Refrigeration System



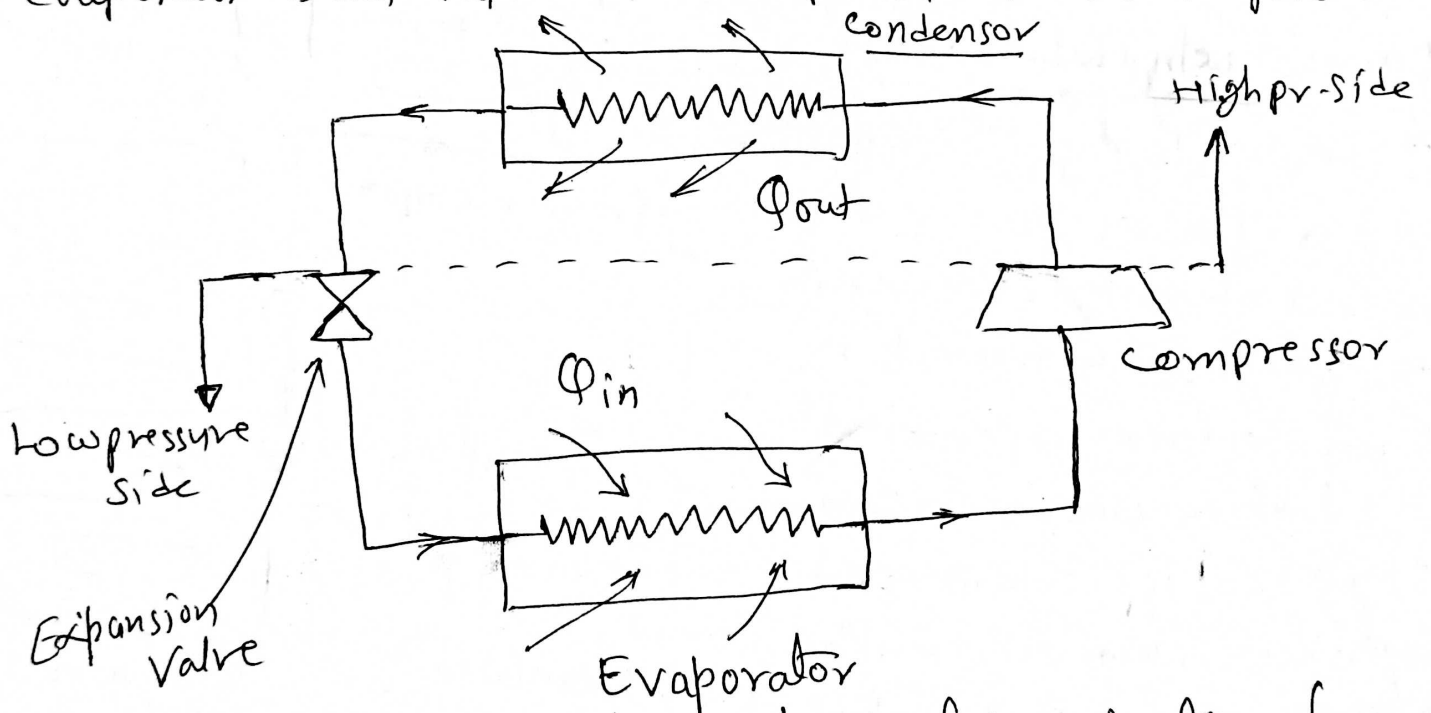
Vapour Absorption Refrigeration System



Method of Refrigeration

- (1) Vapour-Compression Refrigeration system
 - (2) Vapour Absorption Refrigeration system
- ### Vapour Compression Refrigeration system

It is used in refrigeration applications like refrigerator, water cooler, air conditioner and cold storage. The refrigeration effect is produced at evaporator. The refrigerant enters the evaporator as wet vapour at lower pressure and low temperature.

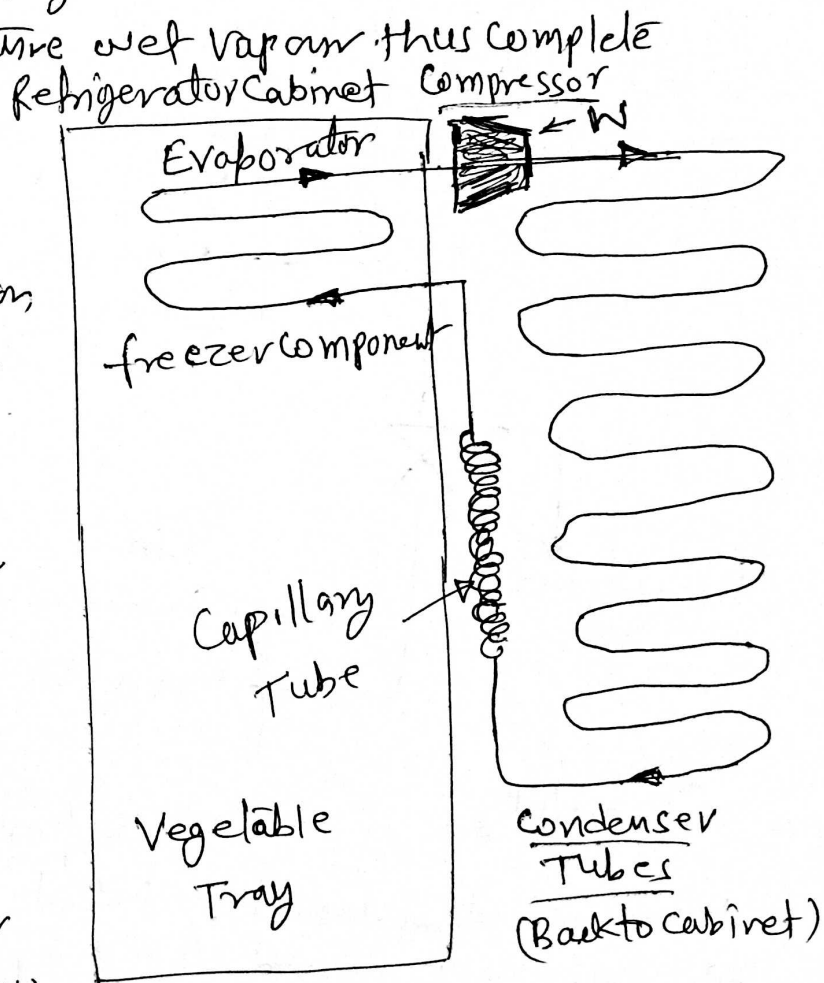


Here it absorbed its latent heat of vaporisation from substance kept around the evaporator thus cooling them. The refrigerant usually comes out of evaporator with phase changed.

to dry saturated or slightly superheated state. Now it enters the compressor which is heart of the system. The refrigerant vapour is compressed to a higher pressure and high temperature as superheated vapour in the compressor. For this power is supplied to the motor connected to the compressor and it constitutes the major running cost of system. The compressed refrigerant vapour is led into the condenser. Generally atmospheric air is blown over the condenser using a fan and it carries away the latent heat from refrigerant vapour. The refrigerant vapour is condensed into high pressure liquid. The liquid refrigerant enters the expansion valve which can be long, spirally wound capillary tube in small refrigerators. The refrigerant is throttled to the low pressure, low temperature wet vapour thus complete cycle.

Domestic Refrigeration System

It belongs to vapour compression system. The evaporator which is coldest part is located in the freezer compartment. A separate door is provided for the freezer where we can store ice, ice-cream and perishable items like mutton, chicken, fish etc. Just below the freezer usually a chiller tray is provided. Further below



and behind the main door there are several compartments with progressive high temperature. The bottom most compartment is for vegetable where a low temperature is not necessary.

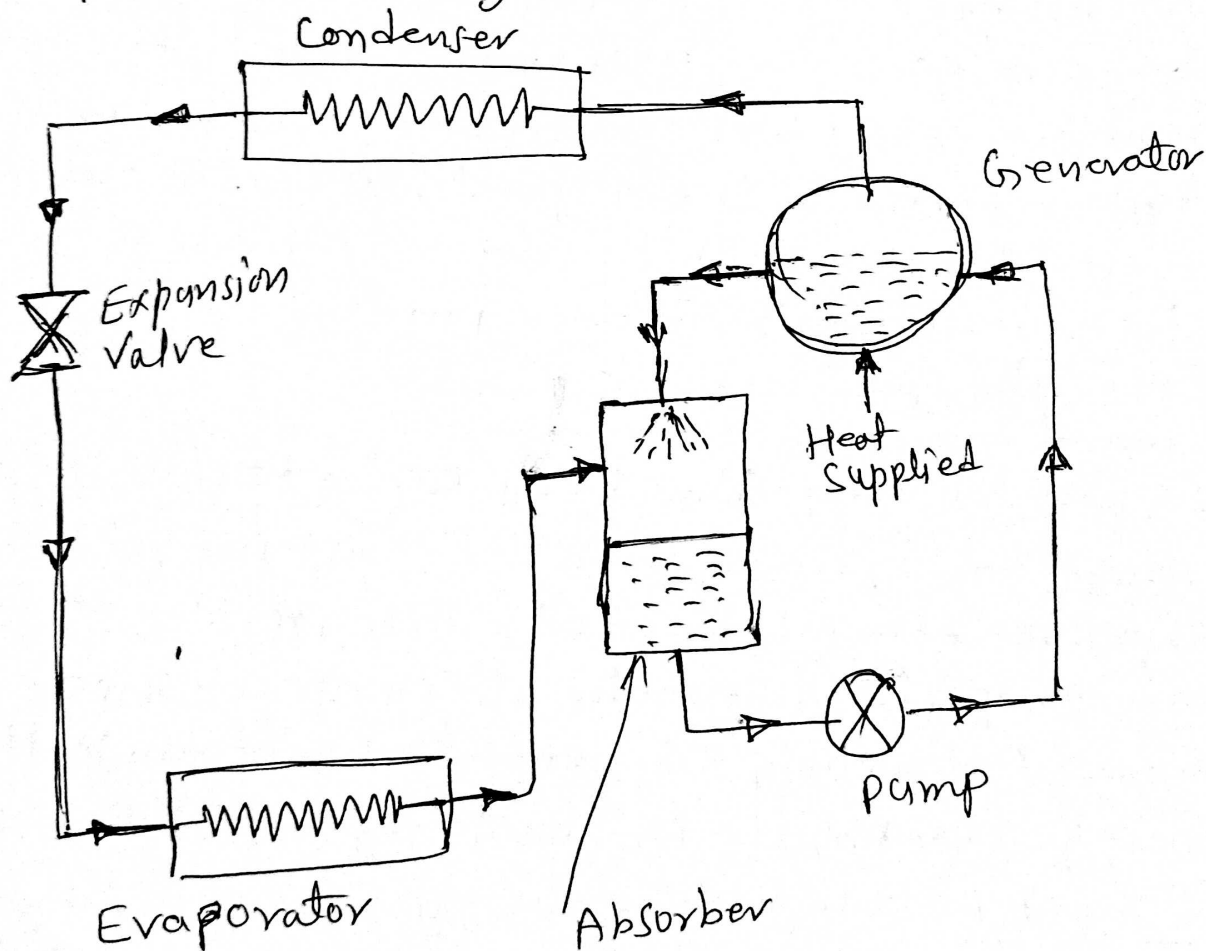
The condensor tubes are kept on the back of refrigerator cabinet. The refrigerant vapour is condensed with the help of surrounding air which rises above by natural convection. In no frost refrigerator the evaporator is located outside the freezer compartment. The cold air is made to flow into the freezer compartment by a fan.

Vapour Absorption Refrigeration System

In this system the low pressure vapour coming out of the evaporator is compressed. The system eliminates the compressor yet produces the compressor effect by combined effects provided by an absorber, a pump and a generator. The pump consumes a lesser amount of electricity. The generator can be run by the heat energy obtained by the burning of any low cost fuel or any heat source including solar energy.

The refrigerant used is ammonia and absorber is water. Ammonia vapour coming out from the evaporator at low pressure is absorbed by water available in the absorber tank resulting in a strong ammonia solution. The pump pumps the solution thereby increasing its pressure and sends the high pressure solution into the generator. In generator solution heated by external

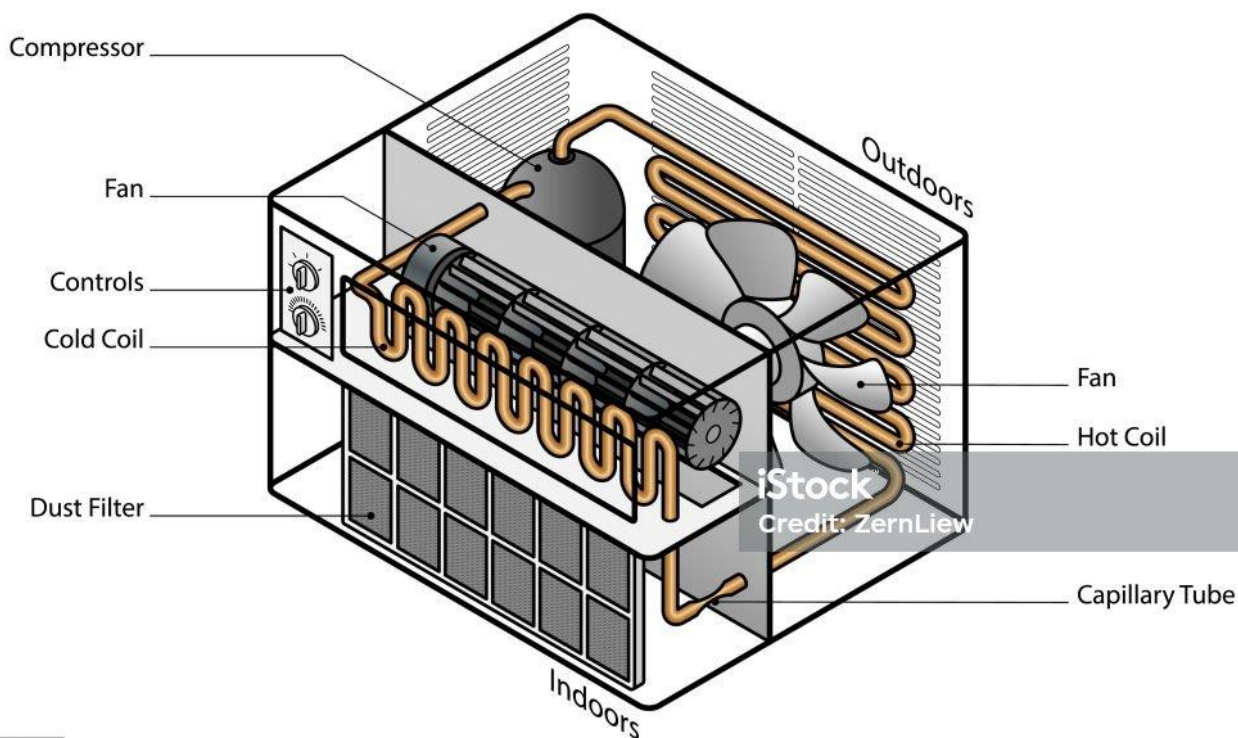
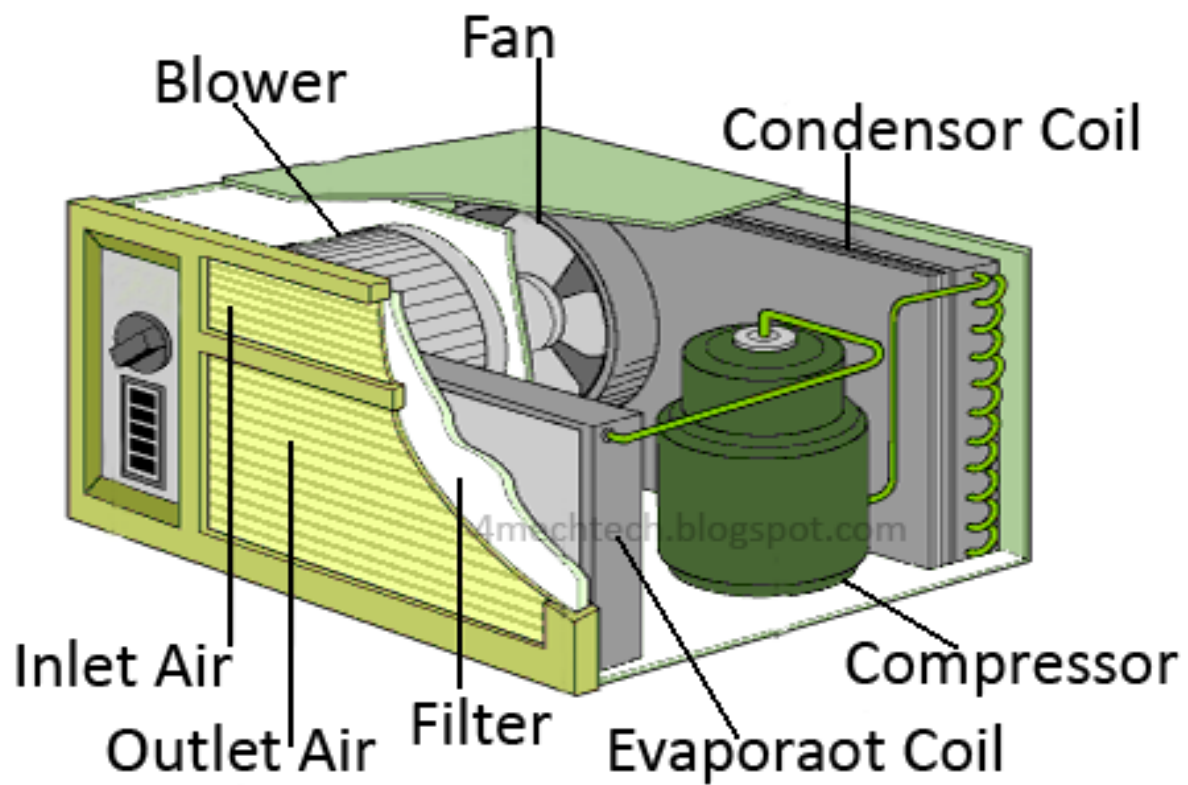
heat supply. Now high pressure ammonia gas is produced which enter the Condenser. The weak ammonia solution available in generator contain more amount of water. It goes back to the absorber tank, where it absorbs the incoming ammonia vapour and becomes a strong solution once again. The processes taking place in the condenser, expansion device and evaporator are similar. The absorber used in vapour absorption refrigeration system should have high affinity for the refrigerant remain in the liquid phase under the operating conditions and should possess high boiling point low specific heat and good chemical stability.



Comparison Between Vapour Compression and vapour Absorption system

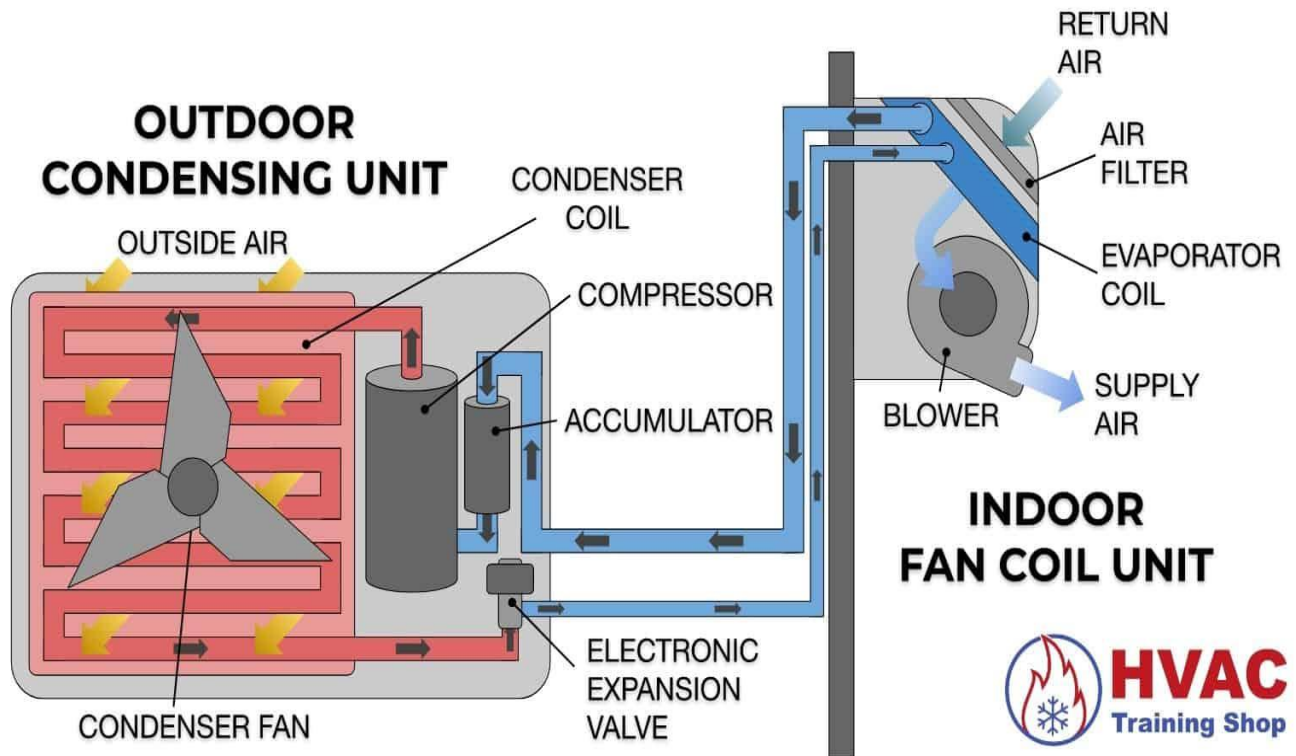
S.No.	Vapour Compression	Vapour Absorption
(1)	Smaller in size	(1) Very large in size for same capacity (Ammonia)
(2)	Refrigerant is Freon-12	(2) Ammonia
(3)	Electric power is used to run the compressor	(3) Heat input can be supplied by a heater or by exhaust steam or even by solar energy.
(4)	COP is higher	(4) Lower
(5)	wear and tear will be more	(5) Less
(6)	system produce noise	(6) silent in operation
(7)	maintenance cost is high	(7) Low

Window Type A.C.

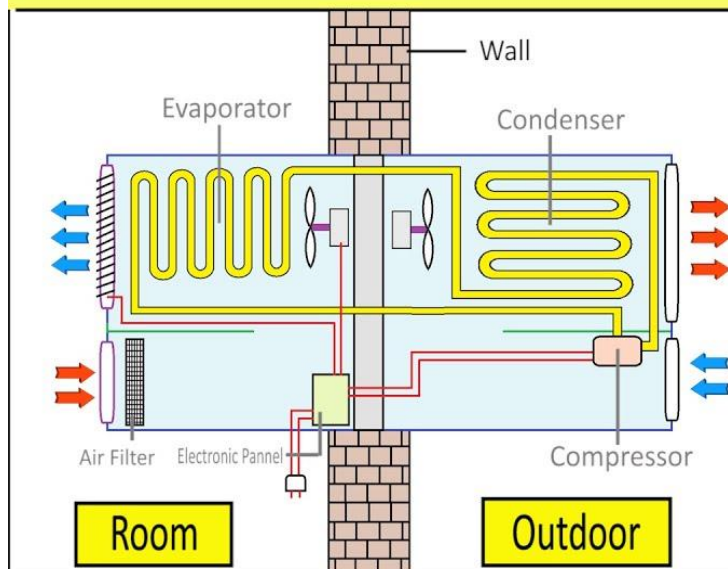


Split Air Conditioner

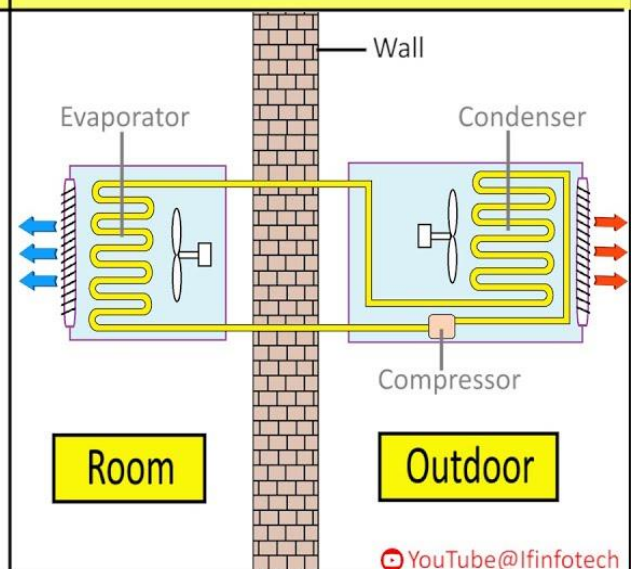
SPLIT SYSTEM AIR CONDITIONER



Window AC



Split AC



Air Conditioning

Air conditioning involves the controlling and maintaining the desired temperature, humidity and direction of flow of air in closed space. Also filtering and cleaning of air is carried out by the air conditioning system.

Air conditioning does not always cooling the air. But during winter air conditioning means heating and humidification.

Application of Air Conditioning

- (1) Air conditioning the houses, hotels and theatres etc
- (2) Hospital, operation theatres and intensive care unit
- (3) for comfort of passengers of cars, buses, trains, ships and aeroplanes.
- (4) Air conditioning become very essential in many industries like textiles, food, printing machine tools etc.

Terminology in Air Conditioning

(1) Dry Air

Air without water vapor or moisture.

(2) Moist Air

It is mixture of air and moisture

(3) Dry Bulb Temperature (DBT)

Actual temperature, measured by a standard mercury thermometer.

(4) Wet Bulb Temperature (WBT)

The temperature measured by mercury thermometer when the bulb cover by wet cloth.

(5) Saturated Air

A mixture of air (dry) along with maximum possible water vapour at DBT.

(6) Relative Humidity

The ratio of the mass of water vapour in a given volume of air at the given temperature to the mass of water vapour present in the same volume under the same temperature of air when it is fully saturated.

Requirements of Comfort Air Conditioning

(1) A Dry Bulb Temperature of 20°C

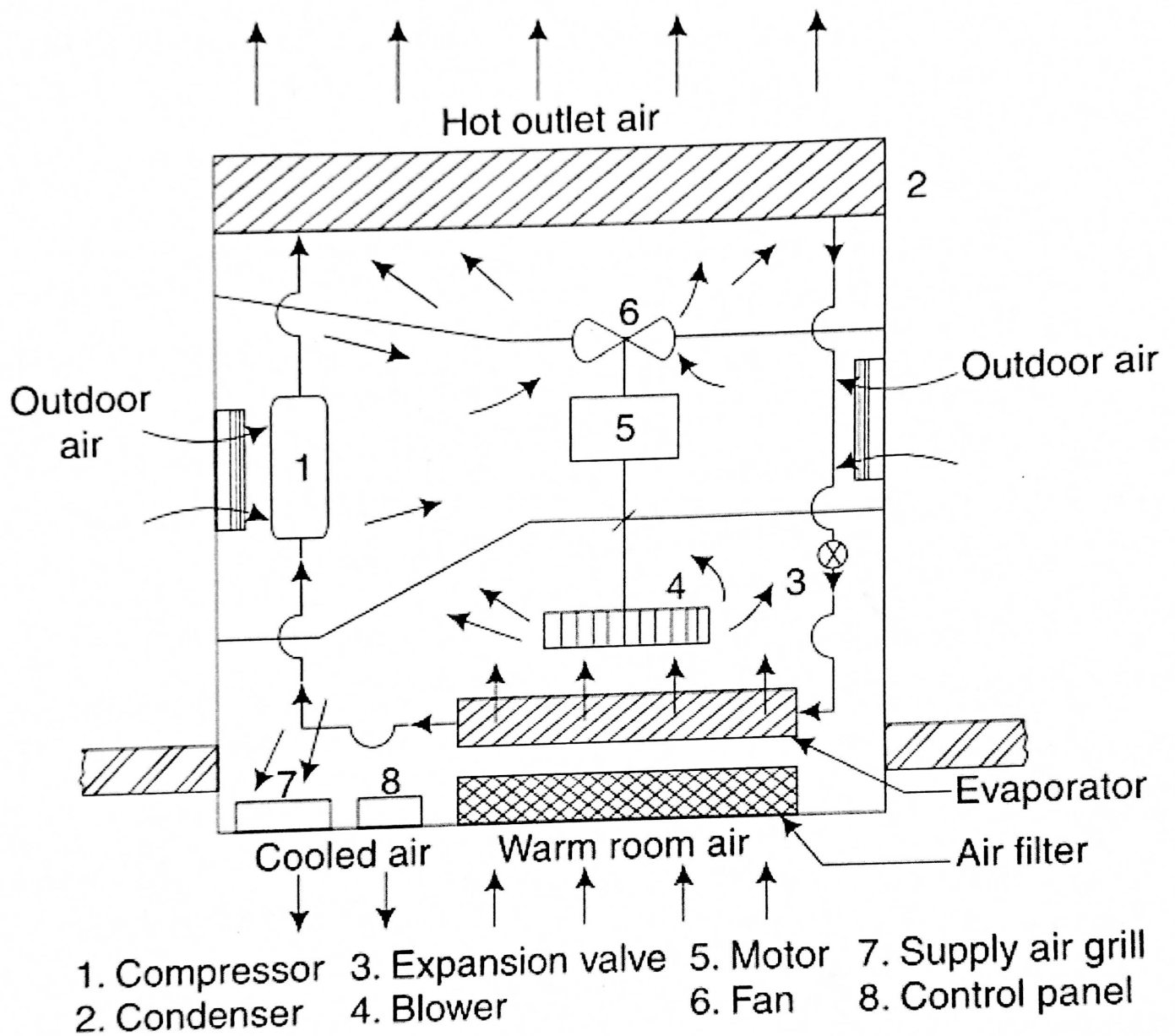
(2) Relative humidity 60% in room

(3) Under Normal condition a person inhales 0.65 m^3 of the oxygen and exhaust (exhales) about 0.2 m^3 of CO_2 in an hour.

(4) A velocity of the range of 8 m/min to 14 m/min .

(5) The conditioned air should be free from dust bacteria, odour and toxic gases.

Window Air Conditioner



Window Air Conditioner

A simple air conditioning system without ducts assembled inside a casing suitable for installation on window or wall opening is called the window air conditioner. The unit consists of Vapour compression system a double shaft motor, a blower, a fan, air filter, supply air grill. Return air grill, fresh air damper, drain tray and control panel. The blower sucks the warm air from the room through air filter and evaporator or cooling coil of the system. Then it deliver cooled and dehumidified air back into the room through the supply air grill. The moisture condensing out the air is passed over the evaporated coil is drained out. The supply air grill adjusted louvers and to change the direction of air flow to ensure uniform distribution of conditioned air inside the room. The controlled conditioned air send into the room. mixes with room air and decreases the temperature and humidity level in room and thereby maintain human comfort. fresh air is admitted through adjustable damper for the purpose of ventilation.

It is available for 0.5 to 3 TR. will operate with 230 Volt and single phase AC supply.

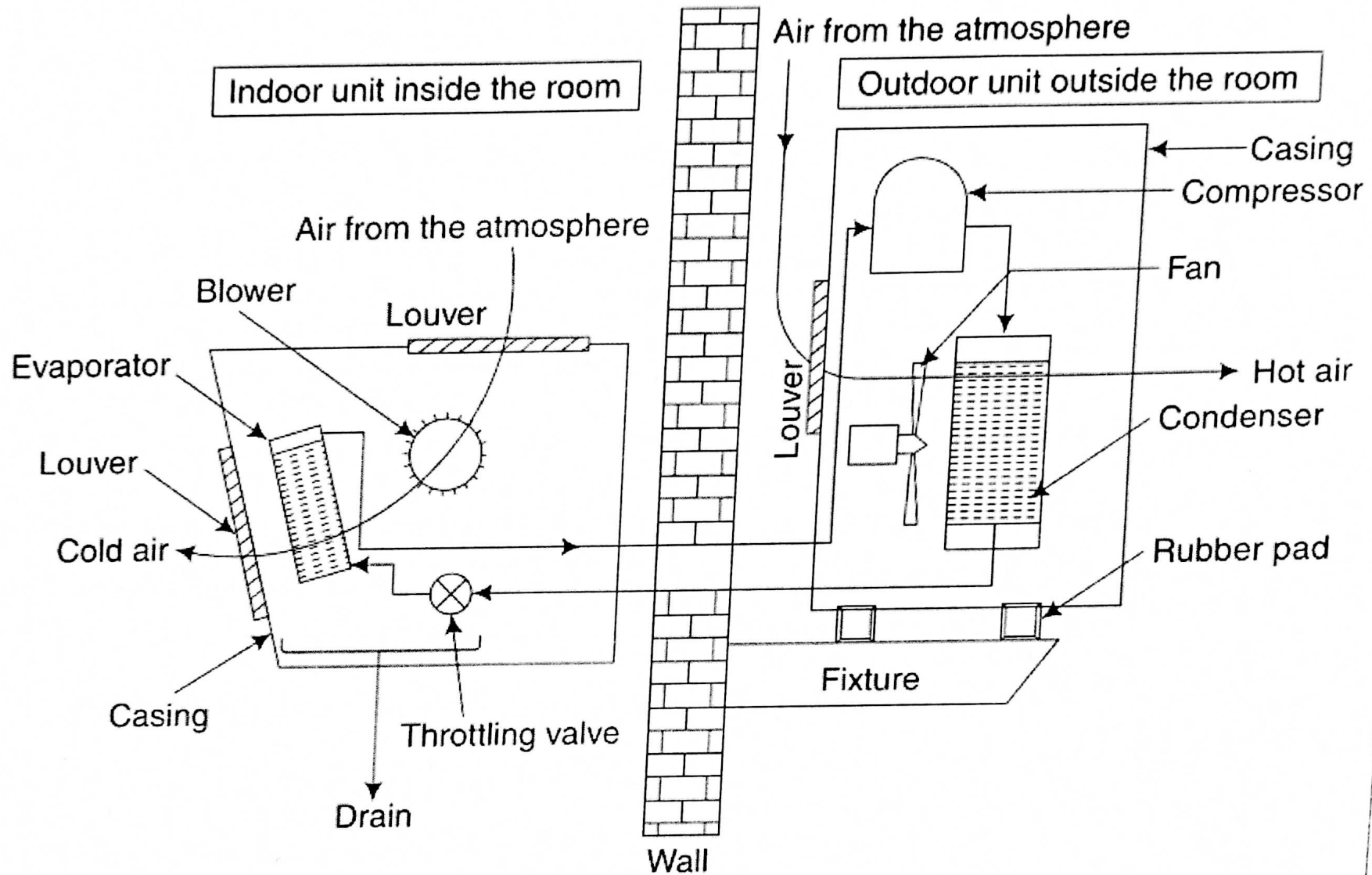
Split Air Conditioner

In split air conditioner the components are placed in two cabinet Indoor cabinet unit and outdoor unit. The Indoor unit placed inside the conditioned room while outdoor unit placed outside the room. They both are connected with the help of tubes in high pressure liquid and low pressure (liquid) vapour would pass through a hole in the wall.

The main components of Indoor unit are blower (B), evaporator (E) throttling valve (Th.V) and drainage (D) system. The high pressure liquid refrigerant from the outdoor unit is allowed to flow from the throttling valve to obtain low pressure vapour refrigerant. The pressure wet vapour refrigerant is then allowed to pass through evaporator. As the blower blow air over the evaporator, low pressure refrigerant obtained from throttling valve is evaporated to produce chillness in surrounding of the evaporator. The Inlet and outlet louvers are placed at suitable location to enable free cold air flow.

The main component of the outdoor unit are the fan (F) condensor (Cond) and compressor (Comp). The low pressure vapour refrigerant from the evaporator of indoor unit is allowed to flow through the compressor to obtain high pressure & high temperature refrigerant. The high pressure and temperature vapour is then allowed to pass through condensor. As the fan blows air over condensor high pressure vapour condensor to form high pressure liquid refrigerant. Air Inlet and outlet louvers are placed at suitable place to enable free air flow. Rubber pad are used to mount the outdoor unit, the working of compressor will produce vibration.

Split Air Conditioner



Central Air Conditioning

This system is used only for heavy loads of about 20 tons or more for large building, star hotels, hospital; cinema etc.

In factory all air conditioner components are assembled at site in a control room and ducts are used to distribute the conditioned air. The duct work should be designed fabricated and erected carefully.

Differentiate Unitary and Central Air Conditioning system

S.No	Unitary Type	Central
(1)	The capital cost is high	(1) Cost is low
(2)	factory assembled	(2) Assembled at site
(3)	located in the space to be conditioned	(3) located away from the conditioned space
(4)	smaller capacity units	(4) larger capacity unit 20 Tons
(5)	No Need for duct	(5) Extensive duct work is essential
(6)	failure in the system will affect the one room only	(6) will affect all the rooms
(7)	Installation charge is less	(7) much Higher

Thermoelectric Cooling

It works on peltier effect. Two dissimilar conductors replace both vapour and liquid phase. The D.C power source

used to pump electrons from one semiconductor to other. In a pair of semiconductor with different characteristics form a thermoelectric cooling module. They are connected electrically in series and thermally in parallel so that two junctions are created. Semiconductor material are of N & P type. They have either more electron than necessary to complete a perfect molecular lattice structure (N type) or don't have enough electron to complete a lattice structure (P type). The extra electron in the N type material and holes left in the P type material are called carriers and they are against that move the heat energy from the cold to hot junction leading to refrigerant effect. These coolers operate in the range of $+60^{\circ}\text{C}$ to -40°C . Good thermoelectric semiconductor material such as bismuth telluride are widely used in these cooling devices.

